

Aqeel Ahmed

Max-Planck-Institut für Kernphysik
Saupfercheckweg 1, 69117 Heidelberg

✉ aqeelhmed@gmail.com

🎓 inspirehep.net/authors/1091402

🌐 aqeelhmed.github.io

☎ +49 1575 277 8443

🆔 0000-0002-2907-2433

🌐 linkedin.com/in/aqeelhmed

EMPLOYMENT	♦ Postdoc Max-Planck-Institut für Kernphysik, Heidelberg, Germany (Parental leave: Jul 2025 – Nov 2025) Oct 2020 – Dec 2025 ♦ Postdoc Vrije Universiteit Brussel (VUB), Brussels, Belgium Oct 2018 – Sep 2020 ♦ Postdoc (MITP & PRISMA Cluster of Excellence) Johannes Gutenberg University, Mainz, Germany Dec 2017 – Sep 2018 ♦ Research Associate University of Warsaw, Warsaw, Poland Apr 2016 – Nov 2017
EDUCATION	♦ Ph.D. Physics University of Warsaw, Warsaw, Poland Visiting Student: University of California, Riverside, USA (1 year) University of California, Davis, USA (6 months) <i>Thesis: “Scalar fields within warped extra-dimension”</i> Advisor: Prof. Bohdan Grzadkowski Dec 2011 – Jan 2016 ♦ M.Sc. & M.Phil Physics Quaid-i-Azam University, Islamabad, Pakistan <i>Thesis: “Analysis of discrete symmetries in b-baryon decays”</i> Advisor: Prof. Riazuddin Feb 2006 – Apr 2010
SUMMARY	My research interests lie in theoretical particle physics with a focus on model building and phenomenological studies of physics beyond the Standard Model. I have contributed to a wide range of research areas, including dark matter, neutral naturalness, extra-dimensional theories, composite Higgs models, flavor physics, baryo/leptogenesis, and early-universe cosmology, resulting in more than 30 peer-reviewed publications, over 950 citations, and an h-index of 14 (June 2026). Beyond research, I have extensive experience coordinating interdisciplinary scientific activities, facilitating international collaborations, organizing workshops and conferences, mentoring students and early-career researchers, and supporting collaborative research programmes.
PROFESSIONAL ACTIVITIES	♦ Mentoring: Co-supervised and mentored 10+ PhD and Master’s students. ♦ Communication/Management: Presented scientific findings at 50+ international conferences and public talks; organized 10+ technical conferences/workshops, managing logistics and content for 200+ attendees. ♦ Representative: Acted as postdoc representative for the <i>be-h consortium Belgium</i>, aligning technical roadmaps and research goals across 5+ international institutions. ♦ Refereeing: Journal of High Energy Physics (JHEP), Physical Review D (PRD), European Physical Journal C (EPJC), and Advances in High Energy Physics
TEACHING	♦ University of Heidelberg (<i>Standard Model of Particle Physics</i>) 2024 ♦ Vrije Universiteit Brussel (<i>Statistical Physics</i>) 2019 ♦ Quaid-i-Azam University Islamabad (<i>QFT, Particle Physics, Group Theory</i>) 2009 – 2011

- PUBLICATIONS [1] A. Ahmed, Z. Chacko, N. Desai, S. Doshi, C. Kilic, S. Najjari and R. P. R. Sudha, “Long-Lived-Particle Signals of a Composite Hidden Sector through the Neutrino Portal,” *JHEP* **05** (2026) 261 [arXiv:2512.09046]
- [2] A. Ahmed, J. P. Garcés and M. Lindner, “Primordial Dirac Leptogenesis,” *Phys. Rev. D* **113** (2026) no.11, L111702 [arXiv:2511.03794]
- [3] A. Ahmed, J. P. Garcés and M. Lindner, “Radiative symmetry breaking with a scale invariant seesaw mechanism,” *Phys. Rev. D* **112** (2025) no.3, 035026 [arXiv:2504.13243].
- [4] A. Ahmed, Z. Chacko, I. Flood, C. Kilic and S. Najjari, “General form of effective operators from hidden sectors,” *JHEP* **05** (2025) 167 [arXiv:2412.15067].
- [5] A. Ahmed, M. Lindner and P. Saake, “Conformal little Higgs models,” *Phys. Rev. D* **109** (2024) no.7, 075041 [arXiv:2309.07845].
- [6] A. Ahmed, Z. Chacko, N. Desai, S. Doshi, C. Kilic and S. Najjari, “Composite Dark Matter and Neutrino Masses from a Light Hidden Sector,” *JHEP* **07** (2024) 260 [arXiv:2305.09719].
- [7] A. Ahmed, B. Grzadkowski and A. Socha, “Higgs boson induced reheating and ultraviolet frozen-in dark matter,” *JHEP* **02** (2023) 196 [arXiv:2207.11218].
- [8] A. Ahmed, B. Grzadkowski and A. Socha, “Higgs Boson-Induced Reheating and Dark Matter Production,” *Symmetry* **14** (2022) no.2, 306.
- [9] A. Ahmed and S. Najjari, “Ultraviolet freeze-in dark matter through the dilaton portal,” *Phys. Rev. D* **107** (2023) no.5, 055020 [arXiv:2112.14261].
- [10] A. Ahmed, B. Grzadkowski and A. Socha, “Implications of time-dependent inflaton decay on reheating and dark matter production,” *Phys. Lett. B* **831** (2022) 137201 [arXiv:2111.06065].
- [11] A. Ahmed, B. Grzadkowski and A. Socha, “Gravitational production of vector dark matter,” *JHEP* **08** (2020) 059 [arXiv:2005.01766].
- [12] A. Ahmed, S. Najjari and C. B. Verhaaren, “A Minimal Model for Neutral Naturalness and pseudo-Nambu-Goldstone Dark Matter,” *JHEP* **06** (2020) 007 [arXiv:2003.08947].
- [13] A. Ahmed, A. Mariotti and S. Najjari, “A light dilaton at the LHC,” *JHEP* **05** (2020) 093 [arXiv:1912.06645].
- [14] A. Ahmed, B. M. Dillon and S. Najjari, “Dilaton portal in strongly interacting twin Higgs models,” *JHEP* **02** (2020) 124 [arXiv:1911.05085].
- [15] A. Ahmed, A. Carmona, J. Castellano-Ruiz, Y. Chung and M. Neubert, “Dynamical origin of fermion bulk masses in a warped extra dimension,” *JHEP* **08** (2019) 045 [arXiv:1905.09833].
- [16] A. Ahmed, “Heavy Higgs of the Twin Higgs Models,” *JHEP* **02** (2018) 048 [arXiv:1711.03107].
- [17] A. Ahmed, M. Duch, B. Grzadkowski and M. Iglicki, “Multi-Component Dark Matter: the vector and fermion case,” *Eur. Phys. J.* **C78**, no.11 (2017) 905 [arXiv:1710.01853].
- [18] A. Ahmed and B. M. Dillon, “Clockwork Goldstone Bosons,” *Phys. Rev.* **D96** no. 11, (2017) 115031 [arXiv:1612.04011].
- [19] A. Ahmed, B. M. Dillon, B. Grzadkowski, J. F. Gunion, Y. Jiang, “Implications of the absence of high-mass radion signals,” *Phys. Rev.* **D95** no. 9, (2017) 095019 [arXiv:1512.05771].
- [20] A. Ahmed, B. Grzadkowski, J. F. Gunion and Y. Jiang, “Higgs Dark Matter from a Warped Extra-Dimension – the truncated-inert-doublet model,” *JHEP* **10** (2015) 033 [arXiv:1504.03706].
- [21] A. Ahmed, L. Dulny and B. Grzadkowski, “Generalized Randall-Sundrum model with a single thick brane,” *Eur. Phys. J.* **C74** (2014) 2862 [arXiv:1312.3577].
- [22] A. Ahmed, B. Grzadkowski and J. Wudka, “Thick-Brane Cosmology,” *JHEP* **04** (2014) 061 [arXiv:1312.3576].
- [23] A. Ahmed and B. Grzadkowski, “Brane modeling in warped extra-dimension,” *JHEP* **01** (2013) 177 [arXiv:1210.6708].
- [24] A. Ahmed, I. Ahmed, M. J. Aslam, M. Junaid, M. A. Paracha and A. Rehman, “Fourth-generation standard model imprints in $B \rightarrow K^* \ell^+ \ell^-$ decays with polarized K^* ,” *Phys. Rev.* **D85** (2012) 034018 [arXiv:1110.4259].

- [25] A. Ahmed, I. Ahmed, M.A. Paracha, M. Junaid, A. Rehman and M.J. Aslam, “Comparative Study of $B_c \rightarrow D_s^* \ell^+ \ell^-$ Decays in Standard Model and Supersymmetric Models”, [arXiv:1108.1058](#).
- [26] I. Ahmed, M.A. Paracha, M. Junaid, A. Ahmed, A. Rehman and M.J. Aslam, “Analysis of $B_c \rightarrow D_s^* \ell^+ \ell^-$ in the Standard Model Beyond Third Generation”, [arXiv:1107.5694](#).
- [27] A. Ahmed, I. Ahmed, M.A. Paracha and A. Rehman, “ $K_1(1270)$ - $K_1(1400)$ mixing and the fourth generation SM effects in $B \rightarrow K_1 \ell^+ \ell^-$ decays”, *Phys. Rev.* **D84** (2011) 033010 [[arXiv:1105.3887](#)].

- PROCEEDINGS
- [28] A. Ahmed, B. Grzadkowski and A. Socha, “Production of Purely Gravitational Vector Dark Matter”, *Acta Phys. Polon.* **B50** (2019) 1809.
 - [29] A. Ahmed, M. Duch, B. Grzadkowski and M. Iglicki, “Vector-Fermion Dark Matter Model”, *Acta Phys. Polon.* **B48** no. 12, (2017) 2405.
 - [30] A. Ahmed, B. Grzadkowski, J.F. Gunion and Y. Jiang, “Higgs Dark Matter from a Warped Extra-Dimension”, *PoS PLANCK 2015* 002 [[arXiv:1510.05722](#)].
 - [31] A. Ahmed, B. Grzadkowski, J.F. Gunion and Y. Jiang, “Radius stabilization and dark matter with a bulk Higgs in warped extra dimension”, *Acta Phys. Polon.* **B46** no. 11, (2015) 2205, [[arXiv:1510.04116](#)].
 - [32] A. Ahmed, L. Dulny and B. Grzadkowski, “Modeling branes in warped extra-dimension”, *Acta Phys. Polon.* **B44** no. 11, (2013) 2381.

SELECTED TALKS

- ◇ *Primordial Dirac Leptogenesis* June 2025
ITP, University of Heidelberg, Heidelberg
- ◇ *Conformal Dark Sectors* May 2023
Pushing the Limits of Theoretical Physics, MITP, Mainz
- ◇ *Dark Side of the Universe* Mar 2023
A Decade of Discoveries in High-Energy Physics, Brussels
- ◇ *Composite Dark Matter and Neutrino Masses from a Light Hidden Sector*
Seminar at the University of Warsaw (Dec 2021) and THEP Mainz (Jan 2022)
- ◇ *Gravitational Production of Vector Dark Matter* Jan 2021
Teilchen-Tee ITP Heidelberg and MPIK Theory Seminar
- ◇ *A Minimal Model for Neutral Naturalness and pNGB Dark Matter* Apr 2020
CERN Theory BSM forum (online)
- ◇ *Dilaton portal to composite twin Higgs* Sep 2019
SCALARS 2019, Warsaw, Poland
- ◇ *Neutral Naturalness at the LHC* Jun 2019
PLANCK 2019, Granada, Spain
- ◇ *A light scalar window at the LHC* Jun 2019
EoS be.h meeting, Brussels, Belgium
- ◇ *Neutral Naturalness at the LHC* Dec 2018
Winter Solstice Meeting, Brussels, Belgium
- ◇ *Clockwork Mechanism: A 4D/5D perspective on BSM model building* Nov 2018
MPIK seminar, Heidelberg, Germany
- ◇ *Exploring the Scalar Sector of the Twin Higgs Models* Dec 2017
SCALARS 2017, Warsaw, Poland
- ◇ *Clockwork Theory: a proposal to generate hierarchies* May 2017
University of Montpellier, Montpellier, France
- ◇ *Multi-component dark matter scenarii: vector-fermion case* May 2017
PLANCK 2017, Warsaw, Poland
- ◇ *Multi-component dark matter: vector-fermion case* Dec 2016
DISRETE 2016, Warsaw, Poland
- ◇ *Clockwork Composite Higgs* Nov 2016
THEP & MITP seminar, Mainz, Germany

	◇ <i>750 GeV resonance: a messenger from warped extra dimensions?</i> Seminar, University of Warsaw, Warsaw, Poland	May 2016
	◇ <i>Higgs-Radion phenomenology at LHC13</i> Niels Bohr Institute, Copenhagen, Denmark	Jan 2016
	◇ <i>Higgs-radion phenomenology and dark matter in warped extra dimensions</i> CP³ – Origins, Odense, Denmark	Jan 2016
	◇ <i>Higgs-Radion Unification in Warped Extra Dimension</i> SCALARS 2015, Warsaw, Poland	Dec 2015
	◇ <i>Radius stabilization and dark matter with a bulk Higgs in RS model</i> XXXIX International Conference of Theoretical Physics, Ustroń, Poland	Sep 2015
	◇ <i>Higgs dark matter from a warped extra-dimension</i> PLANCK 2015, Ioannina, Greece	May 2015
	◇ <i>Colloquium: A case for large extra-dimensions</i> Mimar Sinan University, Istanbul, Turkey	Aug 2014
	◇ <i>Thick Brane Cosmology</i> SUSY 2014, Manchester, UK	Jul 2014
	◇ <i>Cosmology with thick branes</i> PASCOS 2014, Warsaw, Poland	Jun 2014
	◇ <i>Modeling branes in warped extra-dimension with scalar fields</i> SCALARS 2013, Warsaw, Poland	Sep 2013
	◇ <i>Modeling branes in warped extra-dimension</i> XXXVII International Conference of Theoretical Physics, Ustroń, Poland	Sep 2013
	◇ <i>Thick Branes in Warped Extra Dimensions</i> Physics Department Seminar, University of California, Riverside, USA	Oct 2012
SKILLS	◆ <i>Programming:</i> Python, C/C++ , Fortran, Wolfram Mathematica ◆ <i>Data Science:</i> Numpy, Scipy, Scikit-learn, PyTorch, Scikit-HEP, MCMC ◆ <i>Tools:</i> Git, Linux/Bash, LaTeX, Jupyter, Docker, Microsoft Office ◆ <i>Soft Skills:</i> Project Management, Communication, Critical Thinking, Team Collaboration ◆ <i>Languages:</i> English (Fluent), German (Intermediate, B1/B2)	
DISTINCTIONS	◇ International PhD Projects Program Fellowship, Foundation for Polish Science co-financed by the EU Regional Fund	2011 – 2015
	◇ Pro-quality Scholarship, University of Warsaw	2011 – 2012
	◇ Chancellor Gold Medal (M.Phil), Quaid-i-Azam University Islamabad	2010
	◇ Riazuddin Fellowship in Particle Physics (M.Phil)	2008 – 2009
REFERENCES	◆ Prof. Manfred Lindner (Max-Planck-Institut für Kernphysik, Heidelberg) EMAIL: lindner@mpi-hd.mpg.de	PHONE: +49 6221 516 800
	◆ Prof. Bohdan Grzadkowski (University of Warsaw, Warsaw) EMAIL: bohdan.grzadkowski@fuw.edu.pl	PHONE: +48 22 553 2848
	◆ Prof. Zackaria Chacko (University of Maryland) EMAIL: zchacko@umd.edu	PHONE: +1 301 405 1774
	◆ Prof. Matthias Neubert (Johannes Gutenberg University, Mainz) EMAIL: matthias.neubert@uni-mainz.de	PHONE: +49 6131 392 3681